

Dataset Analysis for Decision-Making

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 sicor_municipal_co... .csv
16.7 MB

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.



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To analyze the dataset and address a specific analytical question, I will first load the dataset (`sicor_municipal_contracts.csv`) and examine its structure and key characteristics. Once the dataset is understood, I will proceed to answer the analytical question using Python code.

Step 1: Load and Inspect the Dataset

I will begin by loading the dataset into a pandas DataFrame and inspecting its structure, including column names, data types, and summary statistics.

```
python
1 import pandas as pd
2
3 # Load the dataset
4 file_path = 'sicor_municipal_contracts.csv'
5 df = pd.read_csv(file_path)
6
7 # Display basic information about the dataset
8 print("Dataset Overview:")
9 print(df.info())
10
11 # Display the first few rows of the dataset
12 print("\nFirst 5 Rows of the Dataset:")
13 print(df.head())
14
15 # Display summary statistics for numerical columns
16 print("\nSummary Statistics:")
17 print(df.describe())
```

Step 2: Identify Key Characteristics

From the output of the above code, I will identify:

1. The number of rows and columns.
2. Column names and their data types.
3. Missing values in each column.
4. Summary statistics for numerical columns (e.g., mean, min, max).
5. Unique values in categorical columns.

Step 3: Address the Analytical Question

Once the dataset's characteristics are understood, I will address the specific analytical question. Please provide the question you would like answered using this dataset.

For example, possible analytical questions could include:

- What is the total value of contracts awarded per municipality?
- Which contractor received the highest total value of contracts?
- How many contracts were awarded in each year?

Please provide your specific question, and I will proceed with the analysis and Python code to answer it.



What are the variables that most influence the differentiation between institutions that contribute the most and the least to the total contract value? Define three options to address this issue and select the best one.



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To address the question of identifying the variables that most influence the differentiation between institutions contributing the most and the least to the total contract value, we need to systematically analyze the dataset. Below, I outline three potential approaches to solve this problem, followed by a selection of the best approach based on its feasibility and effectiveness.

Approach 1: Statistical Analysis Using Group Comparisons

Description :

- Divide institutions into two groups: those contributing the most to the total contract value (top contributors) and those contributing the least (bottom contributors).
- Compare the distributions of key variables (e.g., contract size, number of contracts, geographic location, etc.) between these two groups using statistical tests such as t-tests or ANOVA.
- Identify variables with statistically significant differences between the groups.

Pros :

- Simple and interpretable.
- Provides clear insights into which variables differ significantly between top and bottom contributors.

Cons :

- Assumes normality of data for some statistical tests.
- May not capture complex interactions between variables.

Approach 2: Feature Importance Using Machine Learning

Description :

- Create a binary target variable where institutions are labeled as "top contributors" or "bottom contributors" based on their total contract value.
- Train a supervised machine learning model (e.g., Random Forest, Gradient Boosting) to predict this binary target.
- Use feature importance scores from the model to identify the variables that most influence the differentiation between the two groups.

Pros :

- Captures non-linear relationships and interactions between variables.
- Provides a ranking of variable importance directly from the model.

Cons :

- Requires preprocessing and careful handling of categorical variables.
- May be computationally intensive for large datasets.

Approach 3: Clustering Analysis

Description :

- Perform clustering (e.g., k-means or hierarchical clustering) on the dataset using all relevant variables.
- Analyze the clusters to identify groups of institutions that contribute the most and the least.
- Examine the centroids of these clusters to determine which variables distinguish the high-contributing institutions from the low-contributing ones.

Pros :

- Does not require predefined labels (unsupervised learning).
- Can reveal natural groupings in the data.

Cons :

- Interpretability of clusters may be challenging.



- Requires careful selection of the number of clusters and scaling of variables.

Selection of the Best Approach

The best approach depends on the dataset's characteristics and the specific goals of the analysis. Given the question's focus on identifying **variables** that differentiate institutions, **Approach 2 (Feature Importance Using Machine Learning)** is the most suitable. This approach:

1. Directly quantifies the importance of each variable in differentiating between top and bottom contributors.
2. Handles both numerical and categorical variables effectively.
3. Is robust to non-linear relationships and interactions between variables.

Implementation of Approach 2

Below is the Python code to implement Approach 2:

```
python
1 import pandas as pd
2 from sklearn.model_selection import train_test_split
3 from sklearn.ensemble import RandomForestClassifier
4 from sklearn.preprocessing import StandardScaler, LabelEncoder
5
6 # Load the dataset
7 file_path = 'sicor_municipal_contracts.csv'
8 df = pd.read_csv(file_path)
9
10 # Step 1: Aggregate data to calculate total contract value per institution
11 institution_totals = df.groupby('institution_id')['contract_value'].sum().reset_index()
12 institution_totals.rename(columns={'contract_value': 'total_contract_value'}, inplace=True)
13
14 # Step 2: Define top and bottom contributors
15 threshold_top = institution_totals['total_contract_value'].quantile(0.9) # Top 10%
16 threshold_bottom = institution_totals['total_contract_value'].quantile(0.1) # Bottom 10%
17
18 institution_totals['contributor_group'] = 'middle'
19 institution_totals.loc[institution_totals['total_contract_value'] >= threshold_top, 'contributor_group'] = 'top'
20 institution_totals.loc[institution_totals['total_contract_value'] <= threshold_bottom, 'contributor_group'] = 'bottom'
21
22 # Merge with original dataset
23 df = df.merge(institution_totals[['institution_id', 'contributor_group']], on='institution_id', how='left')
24
25 # Step 3: Prepare features and target
26 # Encode categorical variables
27 categorical_cols = df.select_dtypes(include=['object']).columns.tolist()
28 label_encoders = {}
29 for col in categorical_cols:
30     le = LabelEncoder()
31     df[col] = le.fit_transform(df[col].astype(str))
32     label_encoders[col] = le
33
34 # Define features and target
35 X = df.drop(columns=['contributor_group', 'institution_id', 'contract_value'])
36 y = df['contributor_group'].apply(lambda x: 1 if x == 'top' else 0) # Binary target: top vs. bottom
37
38 # Step 4: Train-test split
39 X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
40
41 # Step 5: Train a Random Forest model
42 rf = RandomForestClassifier(random_state=42)
43 rf.fit(X_train, y_train)
44
45 # Step 6: Extract feature importances
46 feature_importances = pd.DataFrame({
47     'feature': X.columns,
48     'importance': rf.feature_importances_
49 }).sort_values(by='importance', ascending=False)
50
51 # Display top influential variables
52 print("Top Variables Influencing Differentiation Between Institutions:")
53 print(feature_importances.head(10))
```

Explanation of the Code

1. **Data Aggregation** : Contract values are aggregated by institution to calculate total contributions.
2. **Group Definition** : Institutions are labeled as "top," "bottom," or "middle" based on their contribution levels.



3. **Feature Engineering** : Categorical variables are encoded, and irrelevant columns are dropped.
4. **Model Training** : A Random Forest classifier is trained to predict whether an institution belongs to the "top contributor" group.
5. **Feature Importance** : The importance of each variable in differentiating top contributors is extracted and displayed.

Deliverables

- **Answer** : The variables that most influence the differentiation between top and bottom contributors will be listed in descending order of importance.
- **Code** : The above Python script processes the dataset and identifies the key variables.

If you have any additional requirements or specific variables to include/exclude, please let me know!

